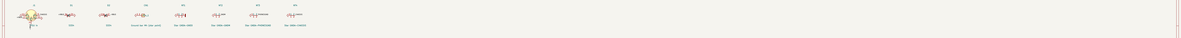
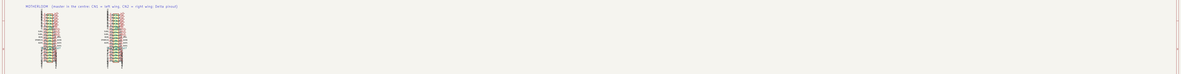
POWER ENTRY: (3-pin XLR from external +/-17 V PSU; reverse-polarity Schottkys; star point)



DISTRIBUTION: (PTC per wing per rail; local branches 4A each; 0P phones; VR mains; HL logic)



MOTHERBOARD: (master is the centre; CN1 = left wing, CN2 = right wing; Delta pinout)



GROUND CANCEL BUFFER: (Feeds every re-inverter summing amps via the raw SIGNCEL line)

MIX L: summing amp (SIGNCEL-referenced), re-inverter, master insert



MIX R: summing amp (SIGNCEL-referenced), re-inverter, master insert



MASTER FADER (dual 100 mm ALPS, pin 1 = GNDA) + post-amps +10 dB

MONO: inverting sum of POST L+R, re-inverter

MAIN OUTPUTS: OPA1134 -> 10x MP in the Sense loop -> mute relay (NO) -> XLR male, pin 2 hot



POWER-ON MUTE: rails > -14 V for ~2 s closes all five relays; either rail low opens them at once



SUBGROUP 1: sum, re-invert, insert, fader + post-amp. DRV134 out, pan to MIX, AFL



SUBGROUP 2: sum, re-invert, insert, fader + post-amp. DRV134 out, pan to MIX, AFL



SUBGROUP 3: sum, re-invert, insert, fader + post-amp. DRV134 out, pan to MIX, AFL



SUBGROUP 4: sum, re-invert, insert, fader + post-amp. DRV134 out, pan to MIX, AFL



AUX MASTER 1: OPA1656 sum (22K || 100p) + re-invert, level, buffer. DRV134 out, AFL



AUX MASTER 2: OPA1656 sum (22K || 100p) + re-invert, level, buffer. DRV134 out, AFL



AUX MASTER 3: OPA1656 sum (22K || 100p) + re-invert, level, buffer. DRV134 out, AFL



AUX MASTER 4: OPA1656 sum (22K || 100p) + re-invert, level, buffer. DRV134 out, AFL



AUX MASTER 5: OPA1656 sum (22K || 100p) + re-invert, level, buffer. DRV134 out, AFL



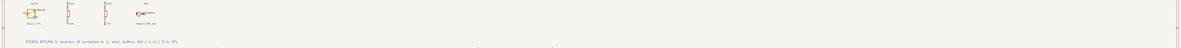
AUX MASTER 6: OPA1656 sum (22K || 100p) + re-invert, level, buffer. DRV134 out, AFL



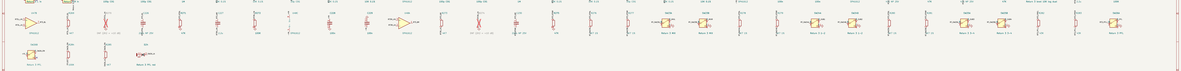
STEREO RETURN 1: receivers (R normalled to L), level, buffers, MIX / 1+2 / 3+4, PFL



STEREO RETURN 2: receivers (R normalled to L), level, buffers, MIX / 1+2 / 3+4, PFL



STEREO RETURN 3: receivers (R normalled to L), level, buffers, MIX / 1+2 / 3+4, PFL



STEREO RETURN 4: receivers (R normalled to L), level, buffers, MIX / 1+2 / 3+4, PFL



CKT INPUT (2-track / DAW) and MIX / CKT monitor select (CW latched = CKT)



PFL / AFL: OPA1656 sum (22K || 100p) + re-invert. SOLO LEVEL, follower



PFL DETECT (1.5 V per press vs 0.7V 4) complementary ramps, rude-solo flasher



PFL SWITCHER: series ±512 (Delta CD2780 sh.1) -> CR5 L/R followers



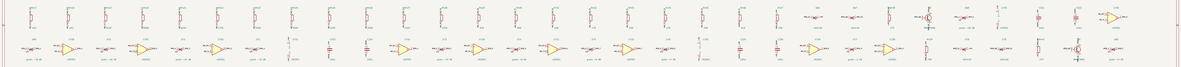
CONTROL ROOM / PHONES: dual ALPS, followers, DRV134, mute relays, TRS outs



HEADPHONES: PHONES dual ALPS, OPA1612 + LM3914N10 composite (+4.9 dB), 2R2 out. PHONES160



REMOTE METER HEADER (for a meter bridge)



METER L: full-wave rectifier, precision peak / average, follower (TL072). 2 dB ladder, 2 x 10-LED strings

METER R: full-wave rectifier, precision peak / average, follower (TL072). 2 dB ladder, 2 x 10-LED strings

Sole concept by Sophie Lille aka Lucy Lacquer
Free to use for non-commercial purposes. Please mention Lucy Lacquer and Lille Audio

Title: Master section + pre console

Rev	Date	By	Change
1.0	2026-04-19	Lucy Lacquer	Initial release